

## Terraform in Azure – Most Asked Interview Questions

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### 1. Basics of Terraform with Azure

#### 1. What is Terraform and why is it used in Azure?

Terraform is an Infrastructure as Code (IaC) tool used to automate provisioning and management of Azure resources in a consistent and repeatable manner.

#### 2. What are Providers? How does AzureRM provider work?

The AzureRM provider is the plugin that allows Terraform to interact with Azure services.

#### 3. What is Terraform State?

Terraform maintains a state file that keeps track of resources created in Azure.

#### 4. Where should Terraform state be stored in Azure?

Azure Storage Account → Blob Container (terraform.tfstate file).

#### 5. What is a Terraform backend?

Backend defines where Terraform state is stored (local, Azure blob, S3, etc.).

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### 2. Terraform State & Locking

#### 6. What is Terraform State Lock?

Prevents multiple users from running Terraform at the same time, avoiding conflicts.

#### 7. How does Azure handle state lock?

Azure uses a **Blob lease** to lock Terraform state.

#### 8. How to unlock a stuck state lock?

Use CLI to break blob lease:

```
az storage blob lease break --container-name tfstate --blob terraform.tfstate
```

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### 3. Terraform Files & Structure

#### 9. What are .tf and .tfvars files?

- .tf: Terraform configuration files
- .tfvars: Variable definitions

#### 10. Structure of common Terraform files

- **main.tf** → Resources
- **variables.tf** → Variable definitions
- **outputs.tf** → Outputs

## 11. What is terraform.tfstate and terraform.tfstate.backup?

State file and its auto-generated backup.

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## 4. Most Common Terraform Commands

### Common Commands:

terraform init

terraform plan

terraform apply

terraform destroy

terraform fmt

terraform validate

terraform refresh      terraform import

terraform taint

terraform state list

terraform state rm

## 12. terraform init

Initializes backend and downloads providers.

## 13. terraform plan

Shows the execution plan.

## 14. terraform apply

Applies the changes.

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## 5. Terraform in Azure DevOps

### 15. How to run Terraform in Azure DevOps pipelines?

Using tasks for:

- Installing Terraform
- Azure CLI login
- Running init/plan/apply

### 16. How to secure backend credentials?

Store secrets in **Azure Key Vault**.

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## 6. Terraform Azure Resources

### 17. How to reference Azure resources in Terraform?

`azurerm_resource_group.rg.name`

### 18. How to define dependencies?

- Implicit: via resource references
  - Explicit: `depends_on = []`
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## 7. Terraform Modules

### 19. What is a Terraform module?

Reusable infrastructure components.

### 20. How to call a module?

```
module "network" {  
  source = "../modules/vnet"  
}
```

### 21. What are Registry modules?

Modules available publicly on Terraform Registry.

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## 8. Provisioners

### 22. What are provisioners?

Scripts used to configure VMs (not recommended).

### 23. remote-exec vs local-exec

- remote-exec: runs inside the VM
  - local-exec: runs on local machine
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## 9. Terraform Import

### 24. How do you import existing Azure resources?

```
terraform import azurerm_resource_group.rg \  
/subscriptions/<id>/resourceGroups/myRG
```

### 25. Does Terraform generate .tf files after import?

No. You must manually write the .tf configuration.

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## **10. Common Terraform Errors**

### **26. Resource already exists**

Fix → Use terraform import.

### **27. Provider version conflict**

Fix → Declare provider version in required\_providers.

### **28. Invalid index / null value**

Occurs due to incorrect variable values.

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## **11. Azure-Specific Questions**

### **29. How to authenticate Terraform to Azure?**

- Azure CLI
- Service Principal
- Managed Identity

### **30. AzureRM provider configuration**

```
provider "azurerm" {  
  features {}  
}
```

### **31. What is the features {} block?**

A mandatory block for AzureRM provider, even if empty.